The integration-time cost of RFI masking for 21 cm intensity-mapping BAO surveys

Dylan Gormley
West Virginia University

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Abstract

Radio-frequency interference (RFI) masking removes data from 21 cm intensity-mapping surveys, but the relevant feasibility question is not whether an interferer is brighter than the BAO signal in a single frame. BAO is inferred statistically from the three-dimensional power spectrum, so what matters is how much cosmological information remains after masking. We develop a first-order mapping from per-channel masked-time fractions to effective thermal noise and usable bandwidth in a Fisher forecast. We consider inverse-variance sample counting and an intentionally unweighted Fourier reduction of heteroscedastic channel noise. Jensen's inequality orders these two reductions within the adopted diagonal-noise model, but they are not claimed to bound every analysis pipeline. Complete frequency excision is represented by a scalar loss of survey volume; the additional radial-mode coupling from spectral gaps and the covariance or bias from residual unmasked RFI are outside this calculation. The method is implemented in the BAONOISE package, with a versioned RadioFisher/CHIME calculation retained as the forecasting backend. The unmasked calculation is checked against the CHIME Overview forecast before any masking model is applied. For the pilot-proxy-derived DTV configuration used as the worked example, the survey-level integration-time penalty is 3.2 per cent, while the time required for a 2 per cent transverse-distance measurement in the most affected redshift bin increases by a factor 1.34. Uniformly masking half of the 470–608 MHz DTV band increases the survey-level time by a factor 1.15. A channel masked 97 per cent of the time is more economically excised than retained in this model. These results quantify the variance cost of masking; a complete detector-threshold calculation also requires a model for the residual RFI that survives the mask.

1 Introduction

Neutral-hydrogen intensity mapping is intended to measure large-scale structure over wide redshift ranges, including baryon-acoustic-oscillation (BAO) distances at $z \sim 1$–2.5 (Seo and Eisenstein, 2007; Bull et al., 2015). CHIME has now detected the cosmological 21 cm signal in cross-correlation with galaxy and quasar catalogues (CHIME Collaboration, 2023), with the Lyα forest (CHIME Collaboration, 2024), and in auto-correlation at $z \sim 1$ (CHIME Collaboration, 2025). These measurements also make the treatment of radio-frequency interference (RFI) a practical cosmology problem rather than only an instrument-characterisation problem.

The 470–608 MHz part of CHIME's 400–800 MHz band contains North American UHF television channels 14–36, each 6 MHz wide (Advanced Television Systems Committee, 2011). For the 21 cm line this corresponds to $z = 1.33$–2.02. Distant transmitters can appear intermittently even at a protected site, particularly under enhanced propagation. The usual response is to flag contaminated time–frequency samples and remove them before mapmaking (e.g. Offringa et al., 2012; CHIME Collaboration, 2022). The CHIME Overview forecast treats persistently contaminated frequency ranges as unavailable, but it does not assign a science cost to the continuum between a clean channel and a completely dead one (CHIME Collaboration, 2022, Fig. 31 and Appendix A).

The question that motivated this work is straightforward: can masking in the DTV band make a BAO measurement impractical even if the detector performs as intended? A comparison between DTV power and a notional "BAO power per frame" does not answer that question. BAO is not a deterministic waveform that must exceed the interference in an individual frame. It is a statistical feature of the three-dimensional power spectrum, recovered by combining many spatial and spectral modes. The useful comparison is therefore between a masking policy and the cosmological information that remains after that policy is applied.

This paper isolates the *variance cost of masking*. Starting from a per-channel masked-time fraction, we (i) map the retained samples to an effective thermal-noise level, (ii) account for completely removed bandwidth with a scalar volume-loss approximation, (iii) propagate both factors through the same Fisher calculation used for the CHIME Overview forecast, and (iv) invert the result to obtain the observing time required for a stated BAO target. The two channel-noise reductions used below are transparent limiting models, not universal bounds on all mapmaking or power-spectrum estimators. Likewise, the scalar treatment of excised bandwidth does not include the full spectral window. Residual RFI that survives the mask is a separate covariance and bias problem and is not absorbed into an "effective integration time."

The software follows the same separation. Simon Foreman's CHIME adaptation of RadioFisher is retained as a versioned scientific backend (Foreman, 2021; Bull et al., 2015); the maintained BAONOISE package owns the channel definitions, exact frequency overlaps, masking policy, Fisher-bank queries, and target calculations. The DTV application is one worked example of that interface rather than a special case in the forecasting code. Throughout the paper we emphasize ratios of required observing time relative to the unmasked calculation. Absolute Fisher times remain conditional on the idealised foreground, calibration, and instrument assumptions, whereas the ratios provide a more direct measure of the additional cost introduced by a specified mask.

2 Forecast framework

We use the Fisher formalism of Bull et al. (2015) in the configuration adopted for the CHIME collaboration's Overview forecast (CHIME Collaboration, 2022, Appendix A and Table 2). Foreman's public `chime-update` branch implements the as-built four-cylinder geometry: four $20\,\mathrm{m} \times 100\,\mathrm{m}$ cylinders on $22\,\mathrm{m}$ centres, 256 dual-polarisation feeds per cylinder at $305\,\mathrm{mm}$ pitch, with $78\,\mathrm{m}$ illuminated, and a baseline density $n(u)$ calculated from that layout (Foreman, 2021). The forecast uses a total system temperature $T_{\mathrm{sys}} = 55\,\mathrm{K}$ including sky contribution, a survey area of $31{,}000\,\mathrm{deg}^2$, the 400–800 MHz band divided into 15 redshift bins, BAO-shift information only, and the idealised foreground setting $\epsilon_{\mathrm{FG}} = 0$. The fiducial parameters are the Planck 2018 final-release CMB best-fit ΛCDM values used in that branch (Planck Collaboration, 2020): $h = 0.6732$, $\Omega_m = 0.3158$, $\sigma_8 = 0.812$, and $\sum m_\nu = 0.06\,\mathrm{eV}$.

For a redshift bin centred at z, the thermal-noise power entering the Fisher integrand can be written as

$$P_N(\mathbf{k}; z) = \frac{T_{\mathrm{sys}}^2(z)\, S_{\mathrm{area}}}{n_{\mathrm{pol}}\, t_{\mathrm{tot}}\, \nu_{21}}\, \mathcal{I}(\mathbf{k}; z). \tag{1}$$

Here $\nu_{21} = 1420.406\,\mathrm{MHz}$, $n_{\mathrm{pol}} = 2$, t_{tot} is the on-sky integration time, and $\mathcal{I}$ contains the baseline density, beam, and channel-window terms that are independent of t_{tot} (CHIME Collaboration, 2022, Eq. A9). The property used here is $P_N \propto t_{\mathrm{tot}}^{-1}$.

Let $F_b^{(0)}(t)$ be the unmasked Fisher matrix for redshift bin b at integration time t. The masking model below returns two scalars for each bin: a retained-time factor $\bar{w}_b$ and a retained-volume factor v_b. The scenario matrix is then

$$F_b(t \,|\, \mathcal{M}) = v_b\, F_b^{(0)}(t\, \bar{w}_b). \tag{2}$$

Equation (2) is exact for the scalar noise-and-volume model adopted in this paper. It is not an exact representation of frequency-gap mode coupling or correlated residual contamination. A bank of $F_b^{(0)}(t)$ matrices therefore evaluates masking scenarios without rerunning the expensive cosmology calculation, while preserving the same approximation used by the direct RadioFisher hooks. Direct and banked evaluations agree to better than 0.2 per cent in σ_A for the tested scenarios.

The BAO observable is the wiggle amplitude $A(z)$, defined through

$$P(k, z) = P_{\mathrm{smooth}}(k, z)\,[1 + A(z)\, f_{\mathrm{bao}}(k)]\,, \tag{3}$$

with fiducial $A_{\mathrm{fid}} = 1$ (Seo and Eisenstein, 2007; Bull et al., 2015). Following the Overview, each redshift bin is analysed independently and marginalised over $\{A, \alpha_\perp, \alpha_\parallel, b\sigma_8, f\sigma_8, \sigma_{\mathrm{NL}}\}$ without external priors. We report the marginalised per-bin BAO-amplitude uncertainty $\sigma_A(z_b)$; smaller values correspond to a more precise measurement. When a signal-to-noise representation is useful, we define

$$\left(\frac{S}{N}\right)_{A,b} = \frac{A_{\mathrm{fid}}}{\sigma_A(z_b)} = \frac{1}{\sigma_A(z_b)}. \tag{4}$$

For statistically independent redshift bins, the survey-level amplitude signal-to-noise ratio is

$$\left(\frac{S}{N}\right)^2_{A,\mathrm{survey}} = \sum_b \left(\frac{S}{N}\right)^2_{A,b} = \sum_b \sigma_A^{-2}(z_b). \tag{5}$$

We use σ_A for per-bin plots and $(S/N)_A$ only when information is combined across bins. We also use the marginalised transverse scale error $\sigma(D_A)/D_A = \sigma(\alpha_\perp)$ and the volume-distance combination

$$\ln D_V = \tfrac{2}{3} \ln \alpha_\perp + \tfrac{1}{3} \ln \alpha_\parallel, \tag{6}$$

which is the quantity plotted in Fig. 31 of the Overview. We do not estimate a 6 MHz BAO signal-to-noise ratio by inverting that figure. Instead, the 6 MHz channel geometry is inserted before the Fisher matrices are marginalised, so the comparison is made in the forecast itself.

3 From a mask table to forecast factors

The model begins with a masked-time fraction $f(\nu)$, so a frequency slice retains a fraction $w(\nu) = 1 - f(\nu)$ of the nominal samples. We assume that flagged samples are removed, that the remaining thermal noise is diagonal in frequency before the line-of-sight transform, and that the mask can be summarised by a frequency-dependent duty fraction. These assumptions exclude residual RFI, time-dependent sky coverage, and the full covariance generated by an irregular spectral window.

A retained slice has noise variance $\sigma^2(\nu) = \sigma_0^2/w(\nu)$. RadioFisher assigns one thermal-noise normalisation to each redshift bin, so this heteroscedastic profile must be reduced to one $\bar{w}_b$. We use two reductions.

Inverse-variance sample counting. If frequency samples are combined with their inverse noise variance, their information adds in proportion to the number of retained samples. The effective time factor is

$$\bar{w}_{b,\mathrm{IV}} = \langle 1 - f \rangle_b, \qquad t_{\mathrm{eff}} = t_{\mathrm{tot}}\, \bar{w}_{b,\mathrm{IV}}. \tag{7}$$

This is the fiducial reduction. Under the stated diagonal-noise assumptions it is the optimal sample-counting result for a mode with uniform spectral weight.

Unweighted Fourier reduction. As a deliberately less adaptive case, suppose the frequency axis is Fourier transformed without compensating for the channel-dependent noise. Each radial mode then inherits the arithmetic mean noise power, giving

$$\bar{w}_{b,\mathrm{F}} = \left\langle \frac{1}{1-f} \right\rangle_b^{-1}. \tag{8}$$

Jensen's inequality guarantees $\bar{w}_{b,\mathrm{F}} \leq \bar{w}_{b,\mathrm{IV}}$, with equality for uniform f. This statement orders the two reductions constructed here; it does not prove that every real estimator lies between them. Calibration errors, correlated noise, nonuniform sky weights, or additional mode cuts can make a real analysis worse than either scalar model. For uniform masking, both reductions give $\bar{w} = 1 - f$, and in the noise-dominated limit the required integration time increases by $1/(1 - f)$.

3.1 Exact channel–bin overlap

A 6 MHz television channel does not generally align with a Fisher redshift bin. For a bin $z \in [z_b^-, z_b^+]$, the corresponding frequency interval is

$$\nu_b^- = \frac{\nu_{21}}{1 + z_b^+}, \qquad \nu_b^+ = \frac{\nu_{21}}{1 + z_b^-}. \tag{9}$$

ATSC channel c occupies

$$\nu_c^- = 470 + 6(c - 14), \qquad \nu_c^+ = \nu_c^- + 6 \quad [\mathrm{MHz}]. \tag{10}$$

Its contribution to bin b is weighted by the exact overlap

$$\Delta\nu_{bc} = \left[\min(\nu_b^+, \nu_c^+) - \max(\nu_b^-, \nu_c^-)\right]_+, \tag{11}$$

where $[x]_+ = \max(x, 0)$. Equations (7) and (8) are evaluated with these overlap lengths, including the clean part of the bin. Channels that cross a redshift-bin boundary are split between the two bins rather than assigned to the nearest centre.

3.2 Excision as a volume approximation

A frequency interval that is removed completely is treated differently from a retained interval with high noise. If a total bandwidth $B_{\mathrm{exc},b}$ is excised from a bin of bandwidth B_b, we use

$$v_b = 1 - \frac{B_{\mathrm{exc},b}}{B_b} \tag{12}$$

as a first-order loss of physical survey volume. The thermal-noise factor $\bar{w}_b$ is then computed over the surviving bandwidth only. This is the same scalar approximation implemented by the RadioFisher hook; it neglects the detailed $k_\parallel$ window produced by the gaps.

The excision threshold is not universal. A useful analytic check can be derived for one contaminated slice occupying a fraction q of a bin. In the noise-dominated, inverse-variance limit, retaining a slice masked by fraction f gives

$$\begin{aligned} R_{\mathrm{keep}} &= \frac{t_{\mathrm{keep}}}{t_0} = \frac{1}{1 - qf}, \\ R_{\mathrm{exc}} &= \frac{t_{\mathrm{exc}}}{t_0} = \frac{1}{\sqrt{1-q}}, \end{aligned} \tag{13}$$

where the square root in R_{exc} follows because Fisher information is linear in volume and proportional to t^2 in the noise-dominated limit. The crossover is

$$f_\star = \frac{1 - \sqrt{1-q}}{q} = \frac{1}{1 + \sqrt{1-q}}. \tag{14}$$

For small q, $f_\star \to 0.5$, which motivates the 0.5 default used for the worked example, but the actual choice should be checked against the science target and the full Fisher bank.

Channel 30 overlaps 3.84 MHz of the $z = 1.40$–1.50 bin, or $q = 0.162$, giving $f_\star = 0.522$ in this analytic limit. The adopted $f = 0.97$ is therefore on the excision side of the crossover. Recovering clean-equivalent depth *within that slice* would require $1/(1 - f) \simeq 33$ times as much integration, but this is not a factor of 33 for the whole bin or survey. In the direct Fisher calculation, channel-30 excision produces a 0.8 per cent survey-level penalty and changes the worst-bin 2 per cent D_A target from 0.318 to 0.363 on-sky years. This direct factor, 1.14, exceeds the analytic $R_{\mathrm{exc}} = 1/\sqrt{1-q} = 1.09$ because Eq. (13) holds in the noise-dominated limit, whereas the bin is partially cosmic-variance limited at this target; volume removed from a partly sample-variance-limited bin cannot be fully recovered with additional integration time. Retaining the channel gives

0.377 years under inverse-variance weighting and 1.984 years under the unweighted-Fourier reduction. For this channel and this target, excision is the lower-cost option under either convention.

4 Software and reproducibility

The software is organised around a narrow boundary between a historical forecast code and the new masking calculation. The RadioFisher fork preserves the CHIME implementation and supplies only the Fisher backend and optional frequency-weight/volume hooks. It is not used as the public interface for mask construction or scenario analysis, and unrelated RadioFisher survey paths do not need to be modernised for this project. The BAONOISE package contains the channel definitions, pilot-proxy ingestion, exact overlap calculation, weighting reductions, excision policy, Fisher-bank interpolation, marginalisation, and required-time inversion.

The expensive step is the clean Fisher bank. For each redshift bin, the full matrix is tabulated on a logarithmic integration-time grid. Building the CHIME bank takes approximately 20 minutes on two cores and is required only when the instrument, cosmology, redshift bins, or forecast assumptions change. Masking scenarios then use Eq. (2), reducing a query to interpolation and matrix marginalisation. A mask table can therefore be evaluated in milliseconds:

```
from baonoise import api

fc = api.load()
mask = {17: 0.333,
        30: 0.970,
        31: 0.235}

api.required_time(fc, mask, target=5.0)
api.required_time(fc, mask,
                  target=5.0, zbin=6)
api.tolerance_curve(fc, band="dtv")
```

The unmasked CHIME calculation is treated as the regression baseline. The repository writes the per-bin $\sigma(D_V)/D_V$ comparison alongside every forecast run. The verification suite also compares bank interpolation with direct Fisher evaluations, checks the expected noise-dominated and cosmic-variance scaling, and evaluates the same scenarios through the direct RadioFisher hooks and the banked path. Unit tests cover channel edges, exact partial overlap, the no-mask identity, uniform masking, excision versus retained-time accounting, and agreement between the piecewise overlap calculation and the frequency-weight hook. Before archival release, the RadioFisher commit, cosmological configuration, redshift edges, baseline file checksum, mask-table checksum, and bank metadata should be pinned in a single versioned reference bundle.

This structure is deliberate. It keeps the original forecast provenance visible while allowing the maintained package to remain small enough to test. The same interface can be reused for another experiment after building a compatible clean Fisher bank. It applies directly to contaminants that can be represented by a retained-time fraction and/or excised frequency intervals; it does not imply that every contaminant has the same residual covariance or bias model.

5 Worked example: CHIME under DTV masking

5.1 Masking scenarios

We use two scenario families.

Uniform stress tests. A common masked fraction f is applied either to the full 470–608 MHz DTV band or to the complete 400–800 MHz survey band. These scenarios isolate the scaling with masked fraction and generate the tolerance curves (Fig. 1).

Pilot-proxy-derived configuration. Table 1 uses the exposure-weighted per-channel fractions from the current pilot-proxy survey product (Gormley and WVU Radio Astronomy Instrumentation Lab, 2026). Quarterly fractions are weighted by the corresponding number of valid frames. Channels 24 and 30 are marked "refused" by that calibration procedure because a clean zero point is not available; the worked example assigns them $f = 0.97$ as an explicit modelling value and excises them under the default $f_{\rm exc} = 0.5$ policy. The other channels retain their measured weighted fractions. This table is the example used in the numerical results, not a universal model for the DTV environment at every site or epoch.

The largest effects occur in three redshift regions: channel 30 and the 31–35 cluster near $z \simeq 1.4$–1.5, channel 24 near $z \simeq 1.65$, and channel 17 near $z \simeq 1.9$. We therefore report both survey-level quantities and targets in the affected bins.

5.2 Unmasked regression against the Overview forecast

The masking calculation is applied only after the unmasked forecast is validated. At the Overview normalisation of 8,760 on-sky hours, the clean per-bin volume-distance errors run from 0.47 per cent at $z = 0.85$, through 0.62 per cent at $z = 1.45$, to 1.04 per cent at $z = 2.43$. They reproduce the percent-level scale and redshift dependence of the ideal-CHIME error bars in

Table 1: Per-channel masked-time fractions f_c used for the pilot-proxy-derived worked example. Channel c occupies the 6 MHz interval defined by Eq. (10). Channels marked "exc." are above the adopted 0.5 threshold and are represented as volume loss.

c	f_c	c	f_c	c	f_c
14	0.012	22	0.010	30	0.970 (exc.)
15	0.010	23	0.017	31	0.235
16	0.014	24	0.970 (exc.)	32	0.140
17	0.333	25	0.014	33	0.101
18	0.011	26	0.040	34	0.023
19	0.012	27	0.020	35	0.138
20	0.011	28	0.014	36	0.012
21	0.013	29	0.013		

Fig. 31 of CHIME Collaboration (2022) using the same instrument, cosmology, binning, information flags, and marginalisation. The corresponding forecast survey-level BAO-amplitude signal-to-noise ratio is 48.45 at one on-sky year. These machine-readable values are used as the no-mask regression product; masking is not allowed to change bins outside its frequency support.

For the pilot-proxy-derived configuration, $\sigma(D_V)/D_V$ at $z = 1.45$ increases from 0.62 to 0.738 per cent, and at $z = 1.65$ from 0.712 to 0.860 per cent. Bins outside the DTV redshift range are unchanged.

5.3 Results

Table 2 reports the survey-level BAO-amplitude signal-to-noise ratio after two on-sky years and the integration-time penalty for a survey-level target of $(S/N)_A = 5$. At this low target the calculation is in the noise-dominated regime, so the same ratios apply closely to other modest survey-level amplitude targets. The aggregate $(S/N)_A = 5$ target is weak for a CHIME-like Fisher forecast because the uncontaminated parts of the band carry substantial information. Per-bin distance and amplitude-S/N targets are therefore more diagnostic of whether the DTV redshifts remain useful (Table 3, Fig. 2).

The pilot-proxy-derived configuration increases the survey-level time by 3.2 per cent under the fiducial reduction. In the most affected bin, the 2 per cent D_A target moves from 0.318 to 0.426 on-sky years, a factor 1.34. The inverse-variance and unweighted-Fourier results differ by only 0.5 per cent for that target after the two near-persistent channels are excised. Most of the cost comes from the excised bandwidth and the channels with $f \gtrsim 0.1$, not from the channels masked at the one-to-few per cent level.

Uniformly masking half of the DTV band gives a 15.1 per cent survey-level penalty. In the $z = 1.40$–1.50 bin, which lies wholly inside the DTV band, the 2 per cent D_A target doubles from 0.318 to 0.636 on-sky years, as expected for uniform 50 per cent masking in the noise-dominated limit. The separate 50 per cent whole-band test gives exactly a factor of two at survey level and serves as an internal scaling check. The 97 per cent DTV stress test still leaves a moderate survey-level amplitude penalty because the clean frequencies outside 470–608 MHz dominate that aggregate target; it should not be read as evidence that the DTV redshift bins remain precise.

Table 2: Survey-level results. $(S/N)_A(2\,\mathrm{yr})$ uses 17,520 on-sky hours. The last column is the time required to reach a survey-level target of $(S/N)_A = 5$, relative to the clean case. Values in parentheses use the unweighted-Fourier reduction where it differs from the fiducial inverse-variance result.

Scenario	$(S/N)_A(2\,\mathrm{yr})$	t/t_{clean}
No masking	63.06	1.000
Pilot-proxy-derived	61.46 (61.40)	1.032 (1.034)
25% of DTV band masked	60.52	1.075
50% of DTV band masked	57.42	1.151
75% of DTV band masked	53.46	1.218
97% of DTV band masked	49.30	1.255
50% of entire band masked	48.45	2.000
ch30 excised	62.51	1.008
ch30 retained	62.68 (59.78)	1.013 (1.064)

Table 3: On-sky time to per-bin targets in the two most affected bins. One on-sky year is defined as 8,760 accumulated hours, matching the Overview forecast normalisation. No calendar-duty factor is assumed.

z bin	Scenario	$(S/N)_{A,b} = 5$	$\sigma(D_A) \leq 2\%$
1.40–1.50	clean	0.177 yr	0.318 yr
1.40–1.50	pilot-proxy	0.238 yr	0.426 yr
1.40–1.50	pilot-proxy (F)	0.239 yr	0.428 yr
1.40–1.50	50% DTV masked	0.353 yr	0.636 yr
1.40–1.50	ch30 retained (F)	1.103 yr	1.984 yr
1.40–1.50	ch30 excised	0.203 yr	0.363 yr
1.60–1.70	clean	0.212 yr	0.416 yr
1.60–1.70	pilot-proxy	0.289 yr	0.554 yr

The channel-30 comparison shows why a single survey-level number is not enough. Excision changes the survey-level time by only 0.8 per cent but costs 16.2 per cent of the $z = 1.40$–1.50 bin's scalar volume. Retaining the channel is only slightly worse under inverse-variance weighting, but it is a factor 6.24 worse than clean for the bin-level D_A target under the unweighted-Fourier reduction. The practical conclusion is narrower than "always excise persistent RFI": for channel 30, at the adopted $f = 0.97$ and for the targets considered here, excision is less expensive than retention.

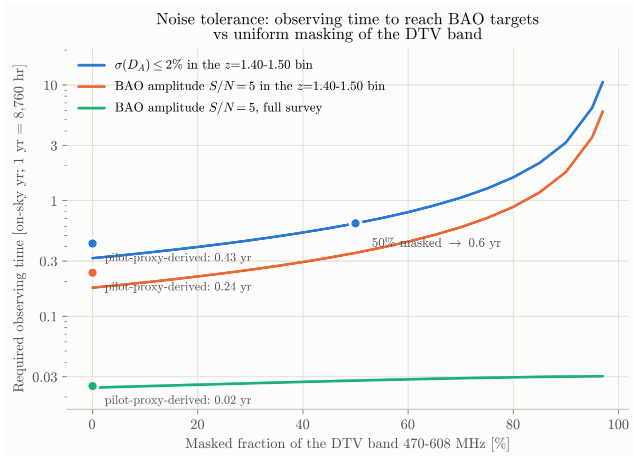

Figure 1: On-sky observing time required for three BAO targets as a function of a uniform masked fraction across 470–608 MHz. The isolated markers at the left edge show the pilot-proxy-derived scenario; their horizontal location is only a plotting convention and is not an effective uniform mask fraction.

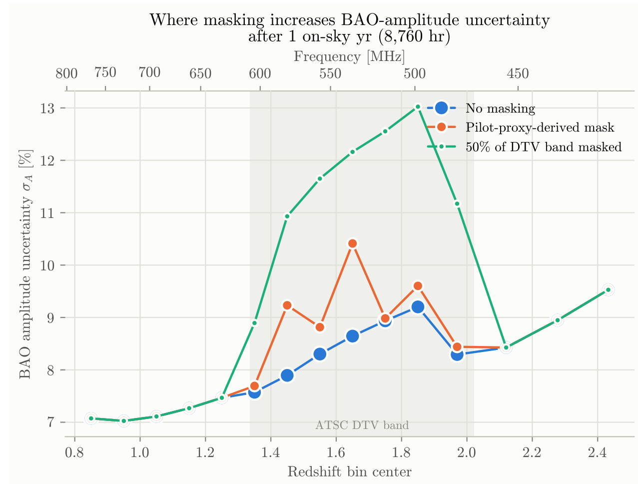

Figure 2: Marginalised per-redshift-bin BAO-amplitude uncertainty σ_A after one on-sky year (8,760 hr), shown as a percentage because $A_{\rm fid} = 1$. Larger values correspond to weaker constraints. Curves are shown for the clean, pilot-proxy-derived, and 50 per cent uniform DTV-band scenarios. The shaded region is the 470–608 MHz DTV band, and the localized increases near $z \simeq 1.45$, 1.65, and 1.9 follow the exact channel–bin overlaps.

5.4 Sensitivity to modelling choices

The main numerical conclusions are stable under four checks. First, once channels 24 and 30 are excised, the two noise reductions give nearly the same answer for the pilot-proxy-derived scenario: the survey penalties are 1.032 and 1.034, and the worst-bin D_A times are 0.426 and 0.428 on-sky years. The large difference appears only when a near-dead slice is deliberately kept in an unweighted transform.

Second, the analysis was repeated with the original five-cylinder CHIME design study in Bull et al. (2015), which differs in geometry, noise model, sky area, cosmology, and parameter treatment. The whole-band 50 per cent test remains exactly a factor of two. The pilot-proxy-derived worst-bin D_A penalty changes from 1.34 to 1.43, while the survey-level 50 per cent DTV penalty changes from 1.15 to 1.28 because the DTV band contributes a different fraction of the total information. These comparisons support using penalty ratios, but they do not make the absolute times instrument independent.

Third, in the Bull-configuration bank, increasing the foreground-residual amplitude from $\epsilon_{\rm FG} = 10^{-6}$ to 10^{-5} changes absolute times by approximately 3 per cent and masking penalties by at most 0.4 per cent. This is a limited sensitivity test, not a substitute for a realistic foreground and calibration covariance.

Fourth, the fiducial cosmology. Rebuilding the CHIME bank with the P-ACT-LB parameters from the Atacama Cosmology Telescope DR6 release (Louis et al., 2025), which combine ACT with *Planck*, CMB lensing, and BAO data, shifts absolute required times by approximately 10 per cent — the clean survey-level $(S/N)_A = 5$ time moves from 211.5 to 189.5 on-sky hours — while changing every masking penalty in Tables 2 and 3 by at most 0.13 per cent. Both fiducial cosmologies ship as named configurations of the tool.

All times in the tables and figures are accumulated on-sky hours, with 8,760 hr labelled as one on-sky year to match the Overview forecast. We do not infer a transferable calendar-duty factor from a selected CHIME analysis data set. Night-only selection, calibration quality cuts, missing days, and RFI masking describe different losses and cannot be collapsed into one survey efficiency without a separate accounting model. Calendar time can be reported once an experiment-specific end-to-end efficiency is specified, but applying such a factor here would risk double-counting the RFI mask whose cost we are trying to isolate.

6 Discussion

What the forecast answers. The calculation answers how much additional integration is required after a specified set of samples has been removed, within the CHIME Fisher model. It can distinguish a small amount of masking spread across many channels from a near-persistent transmitter concentrated in one channel, and it reports the result either for the full survey or for the redshift bins that actually contain the interference. This is the appropriate level at which to ask whether masking alone makes a DTV redshift impractical.

Spectral-window limitations. The scalar volume factor in Eq. (12) does not preserve the exact line-of-sight window. Disconnected frequency support couples $k_\parallel$ modes, can redistribute foreground leakage, and may remove more information than the bandwidth fraction alone suggests. An exact treatment would propagate the mask through the mapmaking or quadratic-estimator covariance, for example in an m-mode framework (Shaw et al., 2014, 2015). The present calculation should therefore be read as a first-order tolerance forecast. The approximation is easiest to defend for small excised fractions and should be revisited when broad or numerous gaps dominate a bin.

Residual contamination. Sub-threshold RFI is not equivalent to missing integration time. A stochastic, decorrelated residual may add covariance; a coherent, cyclostationary, or spectrally structured residual can also bias the recovered BAO parameters. For a detector threshold τ, lowering the threshold generally masks more data while reducing the residual. A complete threshold optimisation would minimise a quantity such as

$$\mathrm{MSE}[\widehat{A};\tau] = \sigma_A^2(\tau) + b_A^2(\tau), \tag{15}$$

where this paper supplies the masking contribution to the variance term and a separate residual model must supply any additional covariance and the bias b_A. The pilot-proxy detector operating curve is the natural input to that companion calculation.

Software scope. The maintained result is the small scenario-to-science-cost interface, not a wholesale rewrite of RadioFisher. Keeping the CHIME forecast backend close to Foreman's validated branch reduces the chance that an unrelated software update changes the reference forecast. New experiments can use the same mask and target machinery after producing their own clean Fisher bank. New contaminants can use the same interface when their first-order effect is sample loss or excised bandwidth; residual-specific physics requires an additional model rather than another scalar hook.

Applications beyond this example. The inputs are deliberately minimal — a clean Fisher bank for the instrument and a per-channel masked-time table for the contaminant — so the calculation is not specific to CHIME or to television. Upcoming and proposed 21 cm surveys occupy similarly contested spectrum, including CHORD at 300–1500 MHz (Vanderlinde et al., 2019), HIRAX at 400–800 MHz under a different regulatory environment (Crichton et al., 2022), and PUMA at 200–1100 MHz (Slosar et al., 2019); each is supported by Fisher-level forecasting from which a clean bank can be built, after which the mask-to-time machinery applies without modification. The contaminant need not be broadcast television: mobile-broadband allocations, aeronautical and radar carriers, or satellite downlinks enter identically whenever their first-order effect is flagged time or excised bandwidth. The same conversion speaks to spectrum management. Protection criteria for radio astronomy are conventionally stated as tolerable percentages of observing time affected by interference (International Telecommunication Union, 2003); the present framework turns such a percentage into the quantity an observatory actually budgets — the additional observing time required to reach a stated science target, survey-wide or in the specific redshift bins at risk. Finally, for detector designers, the tolerance curve supplies the variance half of the threshold optimisation in Eq. (15) for any flagging strategy whose statistics can be summarised per channel.

7 Conclusions

The motivating concern was whether DTV masking by itself could make the corresponding CHIME BAO redshifts unusable. Within the idealised CHIME Overview Fisher model and the scalar mask treatment used here, the answer is no for the pilot-proxy-derived configuration: the survey-level integration time increases by 3.2 per cent, and the most affected 2 per cent D_A target increases by a factor 1.34. A uniform 50 per cent mask across the full DTV band increases the survey-level time by a factor 1.15, although the affected bin-level distance targets double. The distinction matters because clean frequencies can hide severe local losses in a survey-wide S/N summary.

For a near-persistent channel, the correct comparison is between retaining a noisy slice and losing its overlapped volume, not between instantaneous DTV and BAO powers. Channel 30 overlaps 16.2 per cent of the relevant redshift bin. At the adopted $f = 0.97$, excision is less expensive than retention for the targets considered here, particularly for an unweighted spectral transform. The 0.5 excision threshold is a useful default, not a universal constant; Eq. (14) and the Fisher bank provide a direct way to check it for another channel or target.

These conclusions apply to the variance cost of samples that are actually removed. They do not show that residual DTV contamination is harmless, and they do not include the complete mode coupling from spectral gaps. Combining the masking-cost calculation with a threshold-dependent residual covariance and bias model is the next step needed to decide how aggressive an RFI detector should be.

Data and software availability

The analysis package and manuscript source distributed with this draft contain the Fisher banks, the pilot-proxy-derived mask table, machine-readable result tables, and the scripts used to generate the figures; the maintained repository is https://github.com/WVURAIL/bao-noise-tolerance. The RadioFisher backend is available from the author's fork (Gormley, 2026), which is based on Foreman's public CHIME branch (Foreman, 2021). The archival release associated with publication should pin the exact commits and record checksums for the baseline-density file, mask table, cosmological configuration, and Fisher bank.

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